DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
ASSIGNMENT

Name of the Student	Kailash M
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Department	B.Tech – Artificial Intelligence and Data Science (AI&DS)
Subject Code & Subject Name	ECA1012 – Microprocessor and Microcontroller
Title of the Assignment	Constraint Based Problem Solving Assignment

ASSESSMENT DETAILS

Course Outcome	Understand Microprocessor architecture and identify different instruction set and addressing modes for arithmetic, logic, data transfer, and branch operations.
Duration / Maximum Marks	1 Week / 100
Bloom's Taxonomy Level	BTL6 (K6) – Create / Design.
Comments / Recommendation	

ASSESSMENT RUBRIC

Stage No.	Assessment Criterion	Maximum Marks	Obtained Marks
1	Stage 1 – Interpret this System Requirement	10	
2	Stage 2 – Classify and Select Instructions	15	
3	Stage 3 – Design the Operational Logic	15	
4	Stage 4 – Addressing Mode Selection Under Constraints	25	
5	Stage 5 – Develop the 8-bit Assembly Language Program	15	
6	Stage 6 – Test, Verify and Critically Evaluate	20	
TOTAL		**100**	

CODE OF CONDUCT

1. I declare that this assignment is my own work. 2. I have not copied from any other source. 3. I have acknowledged all references used. 4. I will follow academic integrity and SIMATS Engineering ethics.	**STUDENT SIGNATURE** Date: _____	**FACULTY SIGNATURE** Date: _____

ENGINEERING DESIGN REPORT

8-bit Microprocessor-Based Monitoring and Control Unit (Intel 8085 Assembly Language)

ECA1012 – Microprocessor and Microcontroller | CO1 | Kailash M | Reg. No. 192524029

Report flow: Requirement → Architecture → Instruction classification → Operational logic → Addressing-mode selection → Algorithm → ALP design → Execution-cycle analysis → Testing → Engineering decision.

STAGE 1 – Interpret the System Requirement

1.1 Role of the 8-bit processor in the system

The equipment-room unit has one job: read two 8-bit sensor values (temperature and humidity), add them into a single 8-bit “stress index”, compare that index with a stored threshold and publish a status. The Intel 8085 is the **only decision-making element**: it fetches instructions and data from memory, computes with its ALU and writes the status, the arithmetic result, the control bits and the masked result back to memory, where the fan, alarm and humidifier drivers read them.

Major functional blocks: (i) sensor data area in memory (2050H, 2051H), (ii) configuration area (threshold 2052H, mask 2053H), (iii) 8085 CPU (ALU, accumulator and flags, register array, control unit, program counter), (iv) program memory (code at 2000H), (v) output area (3050H, 3051H, 3052H, 3060H).

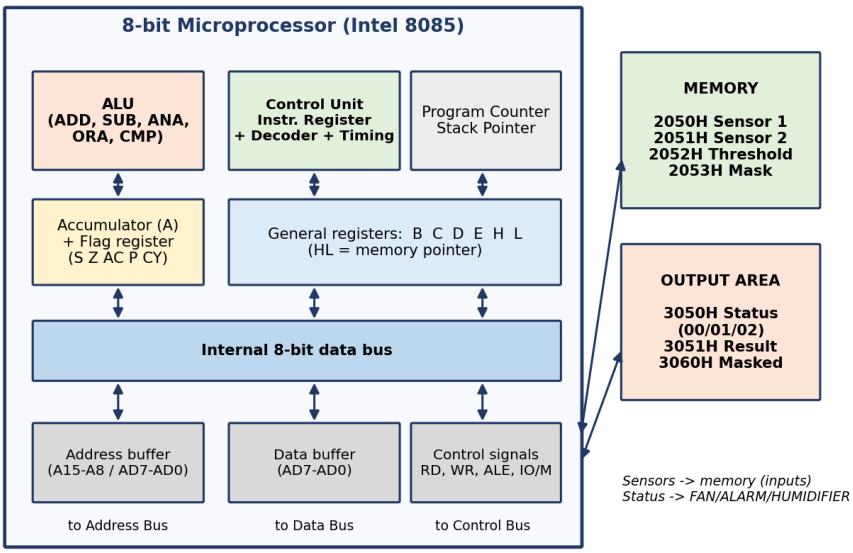


Figure 1: Labelled functional block diagram of the 8-bit processor system

1.2 How the ALU, control unit, registers and buses take part

Block	Participation in this design
Control unit (instruction register, decoder, timing)	Fetches each opcode, decodes it and generates RD, WR, ALE and IO/M signals and the internal enables for every machine cycle. It decides, from the CY flag, whether JC/JNC load a new PC.
Program counter (PC)	Holds the address of the next byte. It runs 2000H → 204AH and is reloaded by JC/JNC/JMP.
ALU	Performs ADD M (S1+S2), ADD D (T+T), CMP D / CMP E / CPI (subtraction for comparison only) and ANA M (bit mask). It updates the flags.
Accumulator (A) and flags	A is the source/destination of every ALU operation and of every STA store. The CY flag drives all three decisions.
Registers B, C, D, E, H, L	B = saved sum, C = status, D = threshold, E = 2T then control byte, HL = memory pointer (see Table 5.4).

Block	Participation in this design
Address bus (16-bit) / data bus (8-bit)	A15–A0 select 2050H…3060H; the 8-bit data bus carries every sensor value, threshold, result and status byte in one transfer.
Control bus	RD for sensor, threshold and mask reads; WR for result, mask-result, status and control stores.

1.3 Why an 8-bit processor is suitable

- Every sensor sample, threshold, mask and status is exactly 8 bits (0–255). One register, one memory cell and one ALU operation handle a whole value, so no multi-byte arithmetic or carry propagation code is required.
- The 16-bit address bus reaches 64 KB, far more than the 75-byte program and ten data bytes need.
- The 8085 needs only a +5 V supply and has an on-chip clock generator, keeping cost and board area low for a simple room controller.
- The task is control-oriented (compare, branch, set a bit), which suits the 8085's flag-based branch instructions.

1.4 Input → processing → decision → output flow

Phase	Operation	Location
Input	Read Sensor 1 and Sensor 2; read threshold T and mask	2050H, 2051H, 2052H, 2053H
Processing	Result = S1 + S2 (8-bit); masked result = Result AND mask	A, B → 3051H, 3060H
Decision	Result < T → NORMAL; T ≤ Result < 2T → WARNING; Result ≥ 2T or overflow → CRITICAL	Flags CY (after CMP/ADD)
Output	Status 00H/01H/02H; control byte (FAN, ALARM, HUMIDIFIER); result; masked result	3050H, 3052H, 3051H, 3060H

1.5 Constraints that shape the design

Constraint	Effect on the design
Instruction constraint (data transfer, arithmetic, logic, branch)	Program must contain at least one of each class: MOV/LDA/STA (transfer), ADD/CMP (arithmetic), ANA/XRA (logic), JC/JNC/JMP (branch).
Register constraint (A, B, C, D, E, H, L only)	Seven registers must hold sum copy, status, threshold, 2T/control and pointer, so values are reused (E is first 2T, then control byte).
Program-memory constraint (≤ 256 bytes)	Favour 1-byte register / register-indirect instructions over 3-byte direct ones where possible; avoid subroutines.
Fixed memory map	Inputs at 2050H–2053H and outputs at 3050H/3051H force direct addressing for the outputs and a pointer (HL) for the inputs.
8-bit data	The sum can overflow beyond FFH; the CY flag must be handled so that the result is never mistaken for a small value.
No simulator / assembler	Opcodes, addresses, flags and results are worked out manually.

1.6 Evolution of microprocessors (two relevant stages)

Stage	Processor	Engineering improvement relevant here
Early	Intel 4004 (1971) – 4-bit, about 2,300 transistors	First single-chip CPU. A 4-bit word needs two operations for one 8-bit sensor byte; this limitation motivated 8-bit CPUs.
8-bit generation	Intel 8085 (1976) – 8-bit data, 16-bit address	One instruction handles an 8-bit sample; single +5 V supply, on-chip clock generator and bus controller and interrupt inputs cut the chip count and cost of a controller.
16-bit generation	Intel 8086 (1978) – 16-bit, 1 MB address	Instruction prefetch queue overlaps fetch and execute, raising speed; shows what a redesign could use for faster monitoring.

System Requirement Statement

The system shall use an Intel 8085 to read two 8-bit sensor values from 2050H and 2051H, add them to an 8-bit result stored at 3051H, apply the mask held at 2053H to give a masked result at 3060H, compare the result with the threshold at 2052H and write the status (00H NORMAL, 01H WARNING, 02H CRITICAL) to 3050H and the FAN/ALARM/HUMIDIFIER control byte to 3052H, using registers A, B, C, D, E, H and L only, at most 256 bytes of program memory, and at least one data-transfer, arithmetic, logic and branch instruction with immediate, register, direct, register-indirect and implied addressing.

STAGE 2 – Classify and Select Instructions

Design rule adopted for the three states (used in all tables below): **NORMAL** if Result < T; **WARNING** if T ≤ Result < 2T; **CRITICAL** if Result ≥ 2T or the 8-bit addition overflowed. With T = 10H the limits are 10H and 20H, which reproduce the three required test cases.

2.1 Instruction-selection table

Required operation	Class	Instruction selected	Addressing mode	Flags / data handling	Reason for selection
Read temperature / humidity	Data transfer	LXI H,2050H; MOV A,M; INX H; ADD M	Immediate (pointer), register indirect	None (MOV, LXI, INX leave flags unchanged); ADD sets all flags	HL pointer reads both sensors in 6 bytes; the second sensor is added straight from memory with no extra load.
Compare with limits	Arithmetic / compare	CMP D; CMP E; CPI 05H	Register; immediate	CMP: CY=1 if A < operand, Z=1 if equal; also S, P, AC; A unchanged	Compares without destroying the result in A.
Increment event counter	Arithmetic	INR M (HL = counter address) or INR B	Register indirect / register	8-bit: FFH→00H wrap; Z, S, P, AC updated; **CY not affected**	Not needed for the core ALP; add LXI H,3070H + INR M (4 bytes) in the CRITICAL branch if an alarm counter is wanted.
Generate / clear control bits	Logic	ANA M (mask); XRA A (clear)	Register indirect; register	ANA/XRA clear CY and AC (ANA sets AC); S, Z, P set from result	ANA with a stored mask clears unwanted bits; XRA A clears A in one byte.
Select NORMAL / WARNING / CRITICAL	Branch / program control	JC, JNC, JMP	Immediate (16-bit target)	JC/JNC test CY only; JMP unconditional	Single flag decides every branch; JMP gives a common exit.
Copy values to display buffer	Data transfer	STA 3050H / 3051H / 3052H / 3060H	Direct	No flags	Output addresses are fixed, so direct addressing is exact and clear.
Optional loop over display buffer	Branch / program control	MVI B,count; MOV M,A; INX H; DCR B; JNZ LOOP	Register indirect; relative-style loop	DCR B sets Z; JNZ loops while Z = 0	Only needed for several buffer bytes; the core ALP writes four fixed bytes so no loop is used, saving bytes and T-states.

2.2 Classification of all instructions used

Class	Instructions in the ALP
Data transfer	LXI H, MOV A,M / MOV B,A / MOV D,M / MOV A,D / MOV E,A / MOV A,B / MOV A,C / MOV A,E, LDA, STA, MVI
Arithmetic	ADD M, ADD D, CMP D, CMP E, CPI, INX H
Logical	ANA M, XRA A
Branch / program control	JC, JNC, JMP
Other control	HLT

2.3 Alternatives rejected under the constraints

Rejected instruction	Why it was rejected
SUB D instead of CMP D	SUB overwrites A, so the sum would have to be restored (extra MOV A,B, +1 byte, +4 T) before the second comparison. CMP gives the same flags without changing A.
JM / JP (sign flag) instead of JC / JNC	Sums above 7FH set the sign flag although the data are unsigned. A sum of 90H would look negative and be misclassified. CY correctly reflects an unsigned “less than”.
CALL / RET for the three outcomes	Each CALL costs 3 bytes + 18 T and RET 10 T, and uses the stack; with only one return point JMP (3 bytes, 10 T) is smaller and faster.
ADI / CPI with constants for the threshold and mask	Hard-codes values that the specification places in memory (2052H, 2053H); changing them would need re-programming. Reading them from memory keeps the program configurable.

STAGE 3 – Design the Operational Logic

3.1 Flowchart

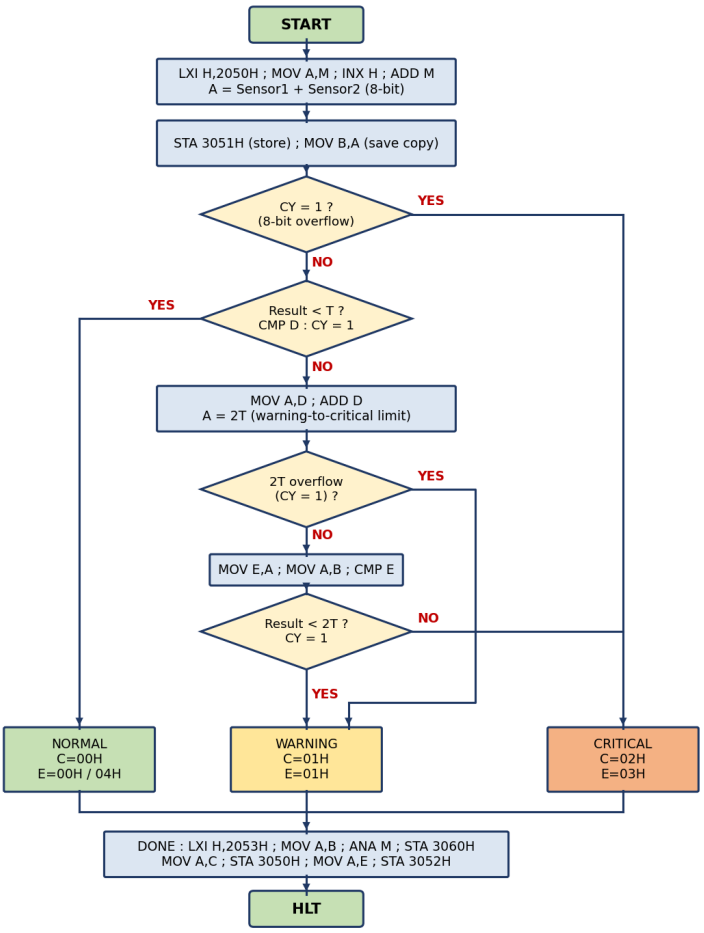


Figure 2: Flowchart of the monitoring and control program (T = threshold, C = status, E = control byte)

3.2 Decision tree

Test (in order)	Condition	Result
1. 8-bit overflow?	CY = 1 after ADD M	CRITICAL (sum wrapped; real value ≥ 100H)
2. Result below threshold?	CY = 1 after CMP D (Result < T)	NORMAL

Test (in order)	Condition	Result
3. 2T overflow?	CY = 1 after ADD D (T ≥ 80H, so 2T ≥ 100H)	WARNING (no 8-bit result can reach 2T)
4. Result below 2T?	CY = 1 after CMP E (Result < 2T)	WARNING; otherwise CRITICAL

3.3 Control-bit definition at 3052H

Bit	Output	Set (1) when	Cleared (0) when
Bit 0 (01H)	FAN	Status is WARNING or CRITICAL (Result ≥ T or overflow)	Status is NORMAL
Bit 1 (02H)	ALARM	Status is CRITICAL (Result ≥ 2T or overflow)	NORMAL or WARNING
Bit 2 (04H)	HUMIDIFIER	Status is NORMAL **and** humidity (2051H) < 05H (dry air)	WARNING, CRITICAL, or humidity ≥ 05H

Resulting control byte: NORMAL = 00H or 04H, WARNING = 01H, CRITICAL = 03H. Status byte at 3050H: 00H, 01H, 02H.

3.4 Flags used by comparison and branch instructions

Flag	Set by	Used by	Meaning here
CY	ADD M, ADD D, CMP D, CMP E, CPI (set when A < operand for compare)	JC, JNC	Overflow of the sum / “less than” in every decision
Z	All ALU operations and CMP (Z=1 if equal)	Not used by a branch	Equality (Result = T) is deliberately merged with WARNING, so no JZ is needed
S, P, AC	ALU operations	Not used	Updated but irrelevant for unsigned classification (S would misclassify values above 7FH)

3.5 How arithmetic, logic and branch instructions cooperate

1. **Arithmetic** – ADD M forms the 8-bit sum and CY reports overflow; CMP D and CMP E turn the sum into CY flags.
2. **Branch** – JC / JNC read CY and route execution to the NORMAL, WARNING or CRITICAL block, each of which loads C (status) and E (control bits).
3. **Logic** – ANA M applies the mask to the stored result; XRA A clears A and E for the NORMAL path.
4. **Transfer** – STA writes the result, masked result, status and control byte to the required addresses.

3.6 Minimum processing steps and eliminated redundancy

Minimum major steps: (1) read S1, (2) add S2, (3) store result, (4) compare with T, (5) compare with 2T, (6) mask, (7) write outputs = **7 major steps**. Redundant operations removed:

- No separate load of Sensor 2: ADD M adds it directly from memory, so no LDA 2051H or temporary register is needed (net saving of 2 bytes and 4 T-states against the direct-mode version).
- One CMP result feeds one JC; no extra equality test (JZ) because Z = 1 is treated as WARNING.
- The threshold is read once into D and reused for both CMP D and ADD D (2T), so the limit 2T is not stored in memory.
- CMP is used instead of SUB, so A never needs to be restored.
- The three outcome blocks share one DONE exit, so the four store instructions exist only once.

Design Decision 1 – Branch strategy

Recommended: a two-level **cascaded conditional branch** on the CY flag (NORMAL test first, then the WARNING/CRITICAL boundary), with the outcome blocks converging on one common exit by JMP/fall-through.

- **Correctness:** CY is the correct unsigned “less than” indicator over the whole range 00H–FFH; overflow is caught before any comparison.
- **Readability:** each outcome has one labelled block (NORM, WARN, CRIT) matching the flowchart one-to-one.
- **Instruction efficiency:** every decision costs one CMP and one 3-byte JC; NORMAL (the most common state) leaves after only two comparisons’ worth of work; there is no CALL/RET stack overhead.

STAGE 4 – Addressing Mode Selection Under Constraints

4.1 Mode-by-mode analysis

Mode	Task	Instruction and operand form	Analysis / suitability
Immediate	Load pointers, status codes and limits	LXI H,2050H (16-bit immediate); MVI C,02H; MVI E,03H; CPI 05H	Best for constants known at design time (pointers, status codes, humidity limit). The threshold and mask are **not** immediate because the specification stores them at 2052H and 2053H.
Register	Intermediate calculations	MOV B,A; MOV A,D; ADD D; CMP D; CMP E; MOV E,A	1-byte, 4-T instructions; the fastest mode. Used to hold the sum (B), threshold (D) and 2T (E) between steps.
Direct	Access 3050H, 3051H, 3052H, 3060H (and 2051H)	STA 3051H; STA 3050H; STA 3052H; STA 3060H; LDA 2051H	3 bytes, 13 T. Ideal for isolated fixed output addresses that are not adjacent, since a pointer would need reloading.
Register indirect	Access 2050H–2053H through HL	MOV A,M; ADD M; MOV D,M; ANA M (HL = 2050H…2053H)	1 byte, 7 T. After one LXI, INX H walks through the four consecutive input bytes cheaply.
Implied	Accumulator / flag / control operation	HLT (no operand); also CMA, STC, CMC available	HLT ends the program using only the opcode. CMA/STC/CMC act on A or CY implicitly; they were not needed because the flag decisions come from CMP.

Design Decision 2 – Selected mode for each data access

Data access	Mode chosen	Cost
Read S1 (2050H), S2 (2051H)	Register indirect via HL	LXI (3 B) + MOV A,M (1 B) + INX H (1 B) + ADD M (1 B) = 6 B, 30 T
Read threshold (2052H) and mask (2053H)	Register indirect via HL (INX H)	MOV D,M and ANA M: 1 B each
Write result, mask result, status, control	Direct (STA)	3 B, 13 T each
Read humidity again (NORMAL path only)	Direct (LDA 2051H)	3 B; HL is still pointing at 2052H, so a pointer move would cost more
Pass values between steps	Register	1 B, 4 T each
Status codes, humidity limit, pointers	Immediate	2–3 B

Cases where another mode works but is less suitable

1. **Reading the sensors with direct addressing** (LDA 2050H; MOV B,A; LDA 2051H; ADD B) works but needs 8 bytes and 34 T-states against 6 bytes and 30 T-states for the HL version, and it needs an extra register (B) just to hold S1.

2. **Masking with an immediate operand** (ANI 0FH, 2 bytes) works and is smaller than ANA M, but it hard-codes the mask and ignores the predefined mask at 2053H; any mask change would need re-programming, so register indirect (ANA M) is more suitable.
3. **Writing outputs through HL** (MOV M,A after LXI H,3050H) works, but the outputs 3050H, 3051H, 3052H and 3060H are not consecutive; reloading HL (LXI, 3 bytes) before each MOV M,A costs 4 bytes per store against 3 bytes for STA.

STAGE 5 – Develop the 8-bit Assembly Language Program

5.1 Memory map

Address	Content	Direction
2050H	Sensor 1 (temperature)	Input
2051H	Sensor 2 (humidity)	Input
2052H	Threshold T	Input (configuration)
2053H	Mask	Input (configuration)
3050H	Status: 00H NORMAL, 01H WARNING, 02H CRITICAL	Output
3051H	8-bit arithmetic result (S1 + S2)	Output
3052H	Control bits: bit0 FAN, bit1 ALARM, bit2 HUMIDIFIER	Output
3060H	Masked result (Result AND mask)	Output
2000H–204AH	Program (75 bytes)	Code

5.2 Algorithm / pseudocode

1. HL ← 2050H; A ← S1; HL ← HL + 1; A ← A + S2 (flags updated).
2. Store A at 3051H; B ← A.
3. If CY = 1 (overflow) → status CRITICAL.
4. HL ← 2052H; D ← T. Compare A with D. If A < T → NORMAL.
5. A ← D + D (2T). If CY = 1 → WARNING. Else E ← 2T, A ← B, compare with E: if A < 2T → WARNING, else CRITICAL.
6. CRITICAL: C ← 02H, E ← 03H. WARNING: C ← 01H, E ← 01H. NORMAL: C ← 00H, E ← 00H; if humidity < 05H then E ← 04H.
7. HL ← 2053H; A ← B AND mask; store at 3060H.
8. Store C at 3050H, store E at 3052H; halt.

5.3 Assembly language program (origin 2000H)

Addr	Machine code	Label	Mnemonic	Comment
2000	21 50 20		LXI H,2050H	HL ← address of Sensor 1 (immediate)
2003	7E		MOV A,M	A ← Sensor 1 (register indirect)
2004	23		INX H	HL ← 2051H
2005	86		ADD M	A ← S1 + S2; CY = overflow
2006	32 51 30		STA 3051H	Store 8-bit result (direct)
2009	47		MOV B,A	B ← result (register)
200A	DA 1E 20		JC CRIT	Overflow → CRITICAL
200D	23		INX H	HL ← 2052H
200E	56		MOV D,M	D ← threshold T
200F	BA		CMP D	Compare result with T (A unchanged)

Addr	Machine code	Label	Mnemonic	Comment
2010	DA 2C 20		JC NORM	Result < T → NORMAL
2013	7A		MOV A,D	A ← T
2014	82		ADD D	A ← 2T
2015	DA 25 20		JC WARN	2T overflowed → WARNING
2018	5F		MOV E,A	E ← 2T
2019	78		MOV A,B	A ← result
201A	BB		CMP E	Compare result with 2T
201B	DA 25 20		JC WARN	Result < 2T → WARNING
201E	0E 02	CRIT:	MVI C,02H	Status ← CRITICAL
2020	1E 03		MVI E,03H	Control ← ALARM + FAN
2022	C3 3A 20		JMP DONE	
2025	0E 01	WARN:	MVI C,01H	Status ← WARNING
2027	1E 01		MVI E,01H	Control ← FAN
2029	C3 3A 20		JMP DONE	
202C	0E 00	NORM:	MVI C,00H	Status ← NORMAL
202E	AF		XRA A	A ← 0, clears flags (logic)
202F	5F		MOV E,A	Control ← 00H (all off)
2030	3A 51 20		LDA 2051H	A ← humidity (direct)
2033	FE 05		CPI 05H	Humidity < 05H ? (CY = 1)
2035	D2 3A 20		JNC DONE	Not dry → skip
2038	1E 04		MVI E,04H	Control ← HUMIDIFIER
203A	21 53 20	DONE:	LXI H,2053H	HL ← mask address
203D	78		MOV A,B	A ← result
203E	A6		ANA M	A ← result AND mask
203F	32 60 30		STA 3060H	Store masked result
2042	79		MOV A,C	A ← status
2043	32 50 30		STA 3050H	Store status
2046	7B		MOV A,E	A ← control byte
2047	32 52 30		STA 3052H	Store control bits
204A	76		HLT	Stop (implied)

Program size: 204BH − 2000H = 75 bytes (4BH), i.e. about 29 % of the 256-byte limit.

5.4 Register-allocation table

Register	Role	Justification
A	Accumulator: S1 + S2, compares, mask, every STA	Only A can be the ALU operand and the source of STA/LDA, so it must carry all arithmetic.
B	Saved 8-bit sum	A is overwritten by MOV A,D and ADD D; B keeps the result for the second compare, the mask and STA.
C	Status code (00H/01H/02H)	Holds the status until the common exit so the three branches can share one set of stores.
D	Threshold T	Read once; reused by CMP D and ADD D (2T).

Register	Role	Justification
E	2T, then control byte	2T is needed only before the decision; E is then reused for control bits, saving a register.
H, L	Memory pointer 2050H → 2053H	One LXI plus INX H reaches all four input bytes with 1-byte instructions.

5.5 Fetch–decode–execute analysis (representative instructions)

Instruction	Fetch	Decode	Execute (machine cycles and T-states)
LXI H,2050H (21 50 20)	M1 (4T): PC → address bus, opcode 21H read into IR, PC + 1	Decoder recognises a 3-byte load of HL; asks for two more bytes	M2 (3T) reads 50H into L; M3 (3T) reads 20H into H. Total 10 T, 3 machine cycles.
MOV A,M (7E)	M1 (4T): opcode 7EH → IR	Source is memory addressed by HL	M2 (3T): HL → address bus, RD, data (S1) → A. Total 7 T.
ADD M (86)	M1 (4T): opcode 86H → IR	ALU add, operand from memory (HL)	M2 (3T): HL → address bus, read S2; ALU adds A + S2 → A; flags updated. Total 7 T.
STA 3051H (32 51 30)	M1 (4T): opcode 32H → IR	3-byte direct store of A	M2 (3T): low byte 51H; M3 (3T): high byte 30H; M4 (3T): address 3051H on bus, WR, A → memory. Total 13 T.
CMP D (BA)	M1 (4T): opcode BAH → IR	ALU subtract A − D, result discarded	Executed within M1: flags set (CY = 1 if A < D, Z = 1 if equal); A unchanged. Total 4 T.
JC NORM (DA 2C 20)	M1 (4T): opcode DAH → IR	Conditional jump on CY; needs 16-bit target	M2 (3T) reads low byte. If CY = 1: M3 (3T) reads high byte and PC ← 202CH (10 T). If CY = 0 the third cycle is skipped, PC is the next instruction (7 T).

5.6 Major instructions and why they were selected

- **ADD M** – adds the second sensor directly from memory; fewest bytes and no extra load.
- **CMP D / CMP E** – flag-only comparison; keeps the sum in A.
- **JC / JNC** – CY is the correct unsigned comparison flag and the only flag the design needs.
- **ANA M** – applies the stored mask (logic class) with no constant hard-coded.
- **STA** – stores to the fixed output addresses by direct addressing.
- **LXI H + INX H** – single pointer walks through all four input bytes.

STAGE 6 – Test, Verify and Critically Evaluate

Test set-up for all cases: threshold 2052H = **10H**, mask 2053H = **0FH**, program origin 2000H, clock 3 MHz (1 T-state = 0.333 µs). The tables list the instructions that change the result, the flags or the stored outputs; pure data moves and pointer increments are omitted for readability.

Test case 1 – Temperature 05H, Humidity 03H (expected NORMAL)

Input memory	2050H	2051H	2052H	2053H
Before execution	05H	03H	10H	0FH

#	Addr	Instruction	A	B	C	D	E	S Z P CY	T total
4	2005H	ADD M	08	00	00	00	00	0 0 0 0	30
5	2006H	STA 3051H	08	00	00	00	00	0 0 0 0	43
7	200AH	JC CRIT (not taken)	08	08	00	00	00	0 0 0 0	54
10	200FH	CMP D	08	08	00	10	00	1 0 0 1	71
11	2010H	JC NORM (taken)	08	08	00	10	00	1 0 0 1	81
12	202CH	MVI C,00H	08	08	00	10	00	1 0 0 1	88
13	202EH	XRA A	00	08	00	10	00	0 1 1 0	92
15	2030H	LDA 2051H	03	08	00	10	00	0 1 1 0	109

#	Addr	Instruction	A	B	C	D	E	S Z P CY	T total
16	2033H	CPI 05H	03	08	00	10	00	1 0 0 1	116
17	2035H	JNC DONE (not taken)	03	08	00	10	00	1 0 0 1	123
18	2038H	MVI E,04H	03	08	00	10	04	1 0 0 1	130
21	203EH	ANA M	08	08	00	10	04	0 0 0 0	151
22	203FH	STA 3060H	08	08	00	10	04	0 0 0 0	164
24	2043H	STA 3050H	00	08	00	10	04	0 0 0 0	181
26	2047H	STA 3052H	04	08	00	10	04	0 0 0 0	198
27	204AH	HLT	04	08	00	10	04	0 0 0 0	203

Working: 05H + 03H = 08H, no carry. 08H < 10H (CMP D: 08 − 10 → borrow, CY = 1) → NORMAL. Humidity 03H < 05H (CPI: CY = 1) → HUMIDIFIER bit set, control = 04H. Masked: 08H AND 0FH = 08H.

Output memory	3050H (status)	3051H (result)	3052H (control)	3060H (masked)
After execution	00H	08H	04H	08H

Total execution: 203 T-states = 67.67 µs at 3 MHz.

Test case 2 – Temperature 08H, Humidity 09H (expected WARNING)

Input memory	2050H	2051H	2052H	2053H
Before execution	08H	09H	10H	0FH

#	Addr	Instruction	A	B	C	D	E	S Z P CY	T total
4	2005H	ADD M	11	00	00	00	00	0 0 1 0	30
5	2006H	STA 3051H	11	00	00	00	00	0 0 1 0	43
7	200AH	JC CRIT (not taken)	11	11	00	00	00	0 0 1 0	54
10	200FH	CMP D	11	11	00	10	00	0 0 0 0	71
11	2010H	JC NORM (not taken)	11	11	00	10	00	0 0 0 0	78
13	2014H	ADD D	20	11	00	10	00	0 0 0 0	86
14	2015H	JC WARN (not taken)	20	11	00	10	00	0 0 0 0	93
17	201AH	CMP E	11	11	00	10	20	1 0 0 1	105
18	201BH	JC WARN (taken)	11	11	00	10	20	1 0 0 1	115
19	2025H	MVI C,01H	11	11	01	10	20	1 0 0 1	122
20	2027H	MVI E,01H	11	11	01	10	01	1 0 0 1	129
24	203EH	ANA M	01	11	01	10	01	0 0 0 0	160
25	203FH	STA 3060H	01	11	01	10	01	0 0 0 0	173
27	2043H	STA 3050H	01	11	01	10	01	0 0 0 0	190
29	2047H	STA 3052H	01	11	01	10	01	0 0 0 0	207
30	204AH	HLT	01	11	01	10	01	0 0 0 0	212

Working: 08H + 09H = 11H (AC = 1, CY = 0). 11H ≥ 10H (CY = 0). 2T = 20H; 11H < 20H (CMP E: CY = 1) → WARNING, control = 01H. Masked: 11H AND 0FH = 01H.

Output memory	3050H (status)	3051H (result)	3052H (control)	3060H (masked)
After execution	01H	11H	01H	01H

Total execution: 212 T-states = 70.67 µs at 3 MHz.

Test case 3 – Temperature 20H, Humidity 15H (expected CRITICAL)

Input memory	2050H	2051H	2052H	2053H
Before execution	20H	15H	10H	0FH

#	Addr	Instruction	A	B	C	D	E	S Z P CY	T total
4	2005H	ADD M	35	00	00	00	00	0 0 1 0	30
5	2006H	STA 3051H	35	00	00	00	00	0 0 1 0	43
7	200AH	JC CRIT (not taken)	35	35	00	00	00	0 0 1 0	54
10	200FH	CMP D	35	35	00	10	00	0 0 0 0	71
11	2010H	JC NORM (not taken)	35	35	00	10	00	0 0 0 0	78
13	2014H	ADD D	20	35	00	10	00	0 0 0 0	86
14	2015H	JC WARN (not taken)	20	35	00	10	00	0 0 0 0	93
17	201AH	CMP E	35	35	00	10	20	0 0 0 0	105
18	201BH	JC WARN (not taken)	35	35	00	10	20	0 0 0 0	112
19	201EH	MVI C,02H	35	35	02	10	20	0 0 0 0	119
20	2020H	MVI E,03H	35	35	02	10	03	0 0 0 0	126
24	203EH	ANA M	05	35	02	10	03	0 0 1 0	157
25	203FH	STA 3060H	05	35	02	10	03	0 0 1 0	170
27	2043H	STA 3050H	02	35	02	10	03	0 0 1 0	187
29	2047H	STA 3052H	03	35	02	10	03	0 0 1 0	204
30	204AH	HLT	03	35	02	10	03	0 0 1 0	209

Working: 20H + 15H = 35H, CY = 0. 35H ≥ 10H and 35H ≥ 20H (CMP E: CY = 0) → CRITICAL, control = 03H. Masked: 35H AND 0FH = 05H.

Output memory	3050H (status)	3051H (result)	3052H (control)	3060H (masked)
After execution	02H	35H	03H	05H

Total execution: 209 T-states = 69.67 µs at 3 MHz.

Additional boundary case 4 – Temperature C0H, Humidity 50H (8-bit overflow, expected CRITICAL)

Input memory	2050H	2051H	2052H	2053H
Before execution	C0H	50H	10H	0FH

#	Addr	Instruction	A	B	C	D	E	S Z P CY	T total
4	2005H	ADD M	10	00	00	00	00	0 0 0 1	30
5	2006H	STA 3051H	10	00	00	00	00	0 0 0 1	43
7	200AH	JC CRIT (taken)	10	10	00	00	00	0 0 0 1	57
8	201EH	MVI C,02H	10	10	02	00	00	0 0 0 1	64
9	2020H	MVI E,03H	10	10	02	00	03	0 0 0 1	71
13	203EH	ANA M	00	10	02	00	03	0 1 1 0	102
14	203FH	STA 3060H	00	10	02	00	03	0 1 1 0	115
16	2043H	STA 3050H	02	10	02	00	03	0 1 1 0	132
18	2047H	STA 3052H	03	10	02	00	03	0 1 1 0	149
19	204AH	HLT	03	10	02	00	03	0 1 1 0	154

Working: C0H + 50H = 110H → A = 10H with CY = 1 (overflow). JC CRIT is taken immediately, so the wrapped result 10H is still stored at 3051H but the status is CRITICAL (a small wrapped value must not look NORMAL). Masked: 10H AND 0FH = 00H.

Output memory	3050H (status)	3051H (result)	3052H (control)	3060H (masked)
After execution	02H	10H	03H	00H

Total execution: 154 T-states = 51.33 µs at 3 MHz.

6.5 Summary of test results

Case	S1 + S2	Result (3051H)	CY after ADD	Status (3050H)	Control (3052H)	Masked (3060H)	Condition
1	05H + 03H	08H	0	00H	04H	08H	NORMAL ✔️
2	08H + 09H	11H	0	01H	01H	01H	WARNING ✔️
3	20H + 15H	35H	0	02H	03H	05H	CRITICAL ✔️
4	C0H + 50H	10H (wrapped)	1	02H	03H	00H	CRITICAL ✔️

6.6 Final verification table

Requirement	Specified	Achieved	Satisfied?	Evidence
Correct sensor data acquisition	Inputs at 2050H, 2051H	S1 read by MOV A,M (HL = 2050H); S2 added by ADD M (HL = 2051H)	Yes	A after ADD in all test traces (08H, 11H, 35H, 10H)
Correct condition classification	NORMAL / WARNING / CRITICAL against threshold 10H	00H, 01H, 02H, 02H written to 3050H	Yes	Tables for cases 1–4
Correct control bits	FAN, ALARM, HUMIDIFIER	3052H = 04H, 01H, 03H, 03H	Yes	Control-bit definition (3.3) and traces
8-bit arithmetic result	Processed result at 3051H	08H, 11H, 35H, 10H (wrap-around flagged by CY)	Yes	Output memory tables
Mask update	Masked result from 3060H onward	3060H = 08H, 01H, 05H, 00H (result AND 0FH)	Yes	ANA M rows in the traces
Instruction-class coverage	Data transfer + arithmetic + logic + branch	MOV/LDA/STA/MVI; ADD/CMP/CPI; ANA/XRA; JC/JNC/JMP	Yes	Listing in 5.3, Table 2.2
Addressing-mode coverage	Immediate, register, direct, register indirect, implied	LXI/MVI/CPI; MOV B,A / CMP D; STA/LDA; MOV A,M / ADD M / ANA M; HLT	Yes	Stage 4 analysis
Program size	≤ 256 bytes	75 bytes (4BH)	Yes	Listing 2000H–204AH
Register constraint	A, B, C, D, E, H, L only	All seven used, none other	Yes	Register-allocation table 5.4
ALP correctness and conciseness	Correct 8-bit ALP	4/4 test cases correct, one common exit, no subroutines	Yes	Summary table 6.5

FINAL ENGINEERING DECISION

Would you use these routines in the system? YES

Criterion	Evidence from the design
Correctness	All four traced cases give the expected status, result, control byte and masked value. The overflow case is handled explicitly; the NORMAL/WARNING/CRITICAL boundaries (T and 2T) are unambiguous.
Instruction efficiency	Longest traced path = 212 T-states (≈ 70.7 µs at 3 MHz); the overflow path is only 154 T-states. Each decision costs one CMP (4 T) and one JC (7/10 T). Reading sensors via HL saved 2 bytes and 4 T versus direct addressing.
Memory usage	75 bytes of the 256-byte budget (29 %), leaving 181 bytes for extensions (event counter, loop, display buffer).
Register usage	All seven allowed registers are used with distinct roles; E is deliberately reused (2T, then control byte). No register is left spare, which is the main constraint on future extensions.
Addressing-mode suitability	Register indirect for the consecutive inputs, direct for the fixed outputs, register for temporaries, immediate for constants and implied for HLT – each choice was the smallest/fastest workable option.
Verification evidence	Manual traces with flags and T-state counts for three required cases plus a boundary case; verification table 6.6 is fully satisfied.

Limitations and recommended improvements

- **Single pass:** the routine runs once. For continuous monitoring wrap it in a loop driven by a timer interrupt (e.g. 8085 RST 7.5) so the CPU is not busy-waiting; this improves reliability and power use.
- **Unsigned data only:** values are treated as unsigned. If T ≥ 80H the 2T limit overflows; the program then reports WARNING for any non-overflow result, which is documented behaviour.
- **Sensor acquisition:** the specification places sensor values in memory. In a real unit an IN instruction from an ADC port would replace the memory read, with no change to the decision logic.
- **Event counter:** adding LXI H,3070H and INR M in the CRITICAL block (4 bytes) would give an alarm counter within the budget.

References

- Gaonkar, R. S., Microprocessor Architecture, Programming, and Applications with the 8085, Penram International.
- Intel Corporation, MCS-80/85 Family User’s Manual (8085 instruction set and timing).